

Technical Report

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Abstract

We investigate learned decoding of a rate-1/2, constraint-length-7 convolutional code under periodic sinusoidal interference, motivated by industrial IoT channels. Three classical baselines are first evaluated: mismatched Viterbi (B1), oracle Viterbi (B2), and FFT-based interference cancellation (B5). At SNR = 5 dB and INR = 5 dB, B1 saturates at BLER = 1.0, B2 achieves 10^{-4} , and B5 achieves 0.35. An end-to-end BiGRU decoder (N1) fails to learn the task — an information bottleneck prevents implicit trellis search — and exceeds the 2.1 MFLOP compute constraint by $3.17\times$. A neural branch metric estimator (N2) resolves this by keeping Viterbi intact and replacing only the branch metric computation with a small BiGRU ($h = 16$, 913,664 MACs). N2 (seed 32) achieves BLER = 2.5×10^{-3} at the same operating point, a gain exceeding five SNR dB over B1 and ≈ 1.2 dB from the oracle by log-linear interpolation. Phase 3 uses an evolutionary algorithm (EA) to search the space of valid 64-state rate-1/2 trellises under two fitness criteria. Under oracle-Viterbi fitness (Phase 3a), the EA improves proxy BLER from 2.1×10^{-2} to 1.1×10^{-2} over 34 generations; full evaluation with 100,000 trials yields BLER = 8×10^{-5} at SNR = 5 dB, matching B2. Under frozen N2 fitness (Phase 3b), the search converges at generation 1 and recovers NASA K=7 exactly, confirming it is optimal among 64-state trellises for this decoder class.

I. Introduction

Classical convolutional decoding via the Viterbi algorithm assumes a precisely known channel model. In AWGN-only environments, mismatched branch metrics computed under this assumption achieve near-optimal block-error-rate (BLER) performance. However, in industrial and IoT settings, narrowband sinusoidal interference from motor drives, power converters, and switching regulators regularly coexists with thermal noise [1], [2]. A standard Viterbi decoder that ignores such interference treats the tonal energy as additional noise, causing severe BLER degradation at moderate interference-to-noise ratios (INR).

This report investigates the following question: *Can a computationally constrained learned component, inserted into an otherwise classical Viterbi decoder, recover most of the performance lost to sinusoidal interference?*

We pursue this question in three phases:

- 1) **Phase 1 — Classical Baselines.** We implement and evaluate the mismatched Viterbi (B1), the oracle Viterbi with perfect interference knowledge (B2), and an FFT-based interference cancellation baseline (B5). These establish the performance floor, ceiling, and a signal-processing mid-point.
- 2) **Phase 2a — End-to-End Neural Decoder (Negative Result).** A bidirectional GRU is trained to map raw received samples directly to information bits. The architecture fails to learn meaningful decoding, plateauing near random-guessing accuracy. Analysis reveals two root causes: an information bottleneck at the pooling layer and the intractability of implicitly representing a 64-state Viterbi search within a small network. The model also violates the 2.1 MFLOP compute constraint.
- 3) **Phase 2b — Neural Branch Metric Estimator.** Motivated by the Phase 2a failure, we decompose the decoding task: a small BiGRU learns local per-step branch metrics capturing interference effects; the Viterbi algorithm performs the global trellis search using those metrics. This architecture succeeds, achieving BLER = 2.5×10^{-3} at SNR = 5 dB, INR = 5 dB, well within the compute budget.
- 4) **Phase 3 — Evolutionary Trellis Search.** We co-design the trellis itself via an evolutionary algorithm (EA) that mutates and evaluates candidate 64-state finite-state machines subject to structural validity constraints. Phase 3a uses oracle Viterbi as the fitness decoder and finds a trellis matching B2

performance under oracle decoding. Phase 3b uses the frozen N2 decoder as fitness and recovers NASA K=7 exactly, validating that the classical trellis is already optimal for this decoder class.

The key thesis is that *problem decomposition* — assigning local channel estimation to a neural network and global sequence search to a classical algorithm — is the critical design principle enabling efficient and effective decoding under structured interference.

II. System Model

A Channel

The discrete-time channel model is

$$r[t] = x[t] + n[t] + i[t], \quad t = 0, 1, \dots, N - 1, \quad (1)$$

where $x[t] \in \{-1, +1\}$ is a BPSK-modulated coded symbol, $n[t] \sim \mathcal{N}(0, \sigma^2)$ is additive white Gaussian noise, and $i[t]$ is a deterministic sinusoidal interferer:

$$i[t] = A \sin\left(\frac{2\pi t}{P} + \varphi\right). \quad (2)$$

The amplitude A is derived from the interference-to-noise ratio (INR):

$$A = \sqrt{2 \cdot 10^{\text{INR}_{\text{dB}}/10} \cdot \sigma^2}, \quad (3)$$

which ensures that the average sinusoidal power equals $10^{\text{INR}_{\text{dB}}/10} \cdot \sigma^2$. The period P is drawn uniformly from $\{8, 9, \dots, 32\}$ samples, the phase φ is drawn uniformly from $[0, 2\pi)$, and the noise variance is $\sigma^2 = 10^{-\text{SNR}_{\text{dB}}/10}$ (with unit symbol energy $E_s = 1$).

B Trellis Code

The encoder is the NASA standard rate-1/2, constraint-length-7 convolutional code with octal generator polynomials $G_1 = 171_8 = (1111001)_2$ and $G_2 = 133_8 = (1011011)_2$ [3]. The constraint length $K_c = 7$ gives memory $m = K_c - 1 = 6$ and $S = 2^m = 64$ trellis states. The free distance is $d_{\text{free}} = 10$.

Encoding processes $K = 256$ information bits. Six tail (zero) bits are appended to terminate the trellis back to state 0, giving $K + m = 262$ total input bits and $N = 2 \times 262 = 524$ coded bits per block. The effective code rate is $R_{\text{eff}} = K/N = 256/524 \approx 0.489$.

C System Parameters

System parameters are given in Table I

D Compute Budget

A compute constraint of 2.1×10^6 multiply-accumulate operations (MACs) per block is enforced for all methods. All profiling uses a consistent MAC-based convention throughout: one MAC counts one multiply-add pair (e.g., a squared difference $(y - x)^2$ is one MAC, not two separate operations). The budget is anchored to the cost of the Viterbi algorithm profiled under this convention.

The Viterbi add-compare-select (ACS) loop uses $N = 512$ coded bits (the $K = 256$ information bits at rate 1/2, excluding tail sections for this budget calculation). There are $T = N/2 = 256$ trellis steps, $S = 64$ destination states, 2 incoming branches per state, and 2 output symbols per branch, each requiring 1 MAC:

$$C_{\text{Viterbi}} = T \times S \times 2 \times 2 = 256 \times 64 \times 4 = 65,536 \text{ MACs}. \quad (4)$$

The 2.1 MFLOP budget therefore allows roughly $32\times$ the cost of the Viterbi ACS loop, providing generous headroom for a neural component while remaining tractable for edge deployment. All methods reported in this work are verified to satisfy this constraint.

TABLE I: Fixed system parameters

Parameter	Value
Modulation	BPSK, $x[t] \in \{-1, +1\}$
Information bits	$K = 256$
Constraint length	$K_c = 7$ (memory $m = 6$)
Trellis states	$S = 64$
Tail bits	$m = 6$ (trellis termination)
Total input bits	$K + m = 262$
Coded block length	$N = 524$
Nominal code rate	$R = 1/2$
Effective code rate	$R_{\text{eff}} = 256/524 \approx 0.489$
Free distance	$d_{\text{free}} = 10$
SNR range	$[0, 10]$ dB
INR range	$[-5, 15]$ dB
Interference period	$P \in [8, 32]$ samples
Interference phase	$\varphi \in [0, 2\pi)$
Eval trials (standard)	10,000 blocks
Target BLER	10^{-3}

III. Phase 1: Classical Baselines

A Mismatched Viterbi (B1)

The mismatched Viterbi decoder assumes an AWGN-only channel. At trellis step t , the branch metric for the transition $s \rightarrow s'$ producing BPSK hypothesis $\mathbf{x}_{s \rightarrow s'} \in \{-1, +1\}^2$ is

$$\gamma_{\text{B1}}(s \rightarrow s', k) = -\frac{\|\mathbf{y}[k] - \mathbf{x}_{s \rightarrow s'}\|^2}{2\sigma^2}, \quad (5)$$

where $\mathbf{y}[k] = (r[2k], r[2k+1])^T$ denotes the received pair at trellis step k (distinct from the scalar sample $r[t]$ in Eq. (1)). Because the interference term $i[t]$ is not modeled, B1 treats sinusoidal energy as noise, causing BLER saturation even at moderate INR.

B Oracle Viterbi (B2)

The oracle Viterbi has full knowledge of A , P , and φ and subtracts the interference before computing branch metrics:

$$\gamma_{\text{B2}}(s \rightarrow s', k) = -\frac{\|\mathbf{y}[k] - \mathbf{x}_{s \rightarrow s'} - \mathbf{i}[k]\|^2}{2\sigma^2}, \quad (6)$$

where $\mathbf{i}[k] = (i[2k], i[2k+1])^T$ is the known interference pair at trellis step k . B2 represents the performance ceiling: it achieves optimal ML decoding under the true channel model and serves as the upper bound for all methods.

C Interference Cancellation + Viterbi (B5)

B5 estimates the interference parameters from the received signal and applies a subtraction prior to mismatched Viterbi decoding:

$$\tilde{\mathbf{y}}[k] = \mathbf{y}[k] - \hat{A} \begin{pmatrix} \sin(2\pi(2k)/\hat{P} + \hat{\varphi}) \\ \sin(2\pi(2k+1)/\hat{P} + \hat{\varphi}) \end{pmatrix}, \quad (7)$$

followed by the B1 branch metric on $\tilde{\mathbf{y}}[k]$.

The estimation pipeline proceeds as follows:

- 1) Compute the periodogram of $r[t]$ via FFT.
- 2) Identify the top-5 magnitude peaks whose frequencies fall within the valid range $[1/P_{\text{max}}, 1/P_{\text{min}}]$.
- 3) For each candidate frequency $\omega_k = 2\pi f_k$, fit $r[t] \approx a_k \cos(\omega_k t) + b_k \sin(\omega_k t) + c$ via least squares:
 $\hat{\mathbf{c}}_k = (\mathbf{A}_k^T \mathbf{A}_k)^{-1} \mathbf{A}_k^T \mathbf{r}.$

- 4) Recover amplitude and phase: $\hat{A}_k = \sqrt{a_k^2 + b_k^2}$, $\hat{\varphi}_k = \text{atan2}(a_k, b_k)$.
- 5) Select the candidate minimizing the residual $\sum_t (r[t] - \hat{A}_k \sin(\omega_k t + \hat{\varphi}_k))^2$.

B5 demonstrates the limit of signal-processing-only interference mitigation.

D Compute Cost

TABLE II: Phase 1 compute cost (per block, 10,000-trial latency median)

Method	FLOPs	Latency (ms)	% of 2.1M budget
B1 Mismatched Viterbi	65,536	0.23	3.1%
B2 Oracle Viterbi	65,536	0.27	3.1%
B5 IC + Viterbi	94,720	0.93	4.5%

Table II summarizes the compute cost of all three classical baselines what was calculated using `thop`. All consume less than 5% of the 2.1 MFLOP budget, leaving substantial headroom for any neural component.

E Results

E1 BLER vs. SNR (INR = 5 dB)

Table III and Fig. 1 show the BLER vs. SNR performance of all three baselines at INR = 5 dB.

TABLE III: BLER vs. SNR at INR = 5 dB (10,000 trials per point)

SNR (dB)	B1	B2	B5
0	1.000	9.46×10^{-1}	9.93×10^{-1}
2	1.000	1.50×10^{-1}	7.69×10^{-1}
4	1.000	1.60×10^{-3}	4.79×10^{-1}
5	1.000	1.00×10^{-4}	3.50×10^{-1}
6	9.79×10^{-1}	$< 10^{-4}$	2.13×10^{-1}
8	1.02×10^{-1}	$< 10^{-4}$	2.65×10^{-2}
10	1.00×10^{-4}	$< 10^{-4}$	1.00×10^{-4}

E2 BLER vs. INR (SNR = 5 dB)

Table IV and Fig. 2 show the BLER vs. INR performance at SNR = 5 dB across the full INR range.

TABLE IV: BLER vs. INR at SNR = 5 dB (10,000 trials per point)

INR (dB)	B1	B2	B5
-5	1.7×10^{-3}	1.0×10^{-4}	3.0×10^{-4}
0	1.14×10^{-1}	1.0×10^{-4}	2.49×10^{-2}
5	1.000	1.0×10^{-4}	3.50×10^{-1}
10	1.000	1.0×10^{-4}	6.73×10^{-1}
15	1.000	1.0×10^{-4}	8.20×10^{-1}

The key observation is that at SNR = 5 dB the interference becomes catastrophic for B1 at INR $\approx$ 5 dB, and B5's estimation error limits its benefit beyond INR $\approx$ 0 dB. The $\approx$ 17 dB gap between B1 and B2 at INR = 5 dB defines the opportunity for a learned approach.

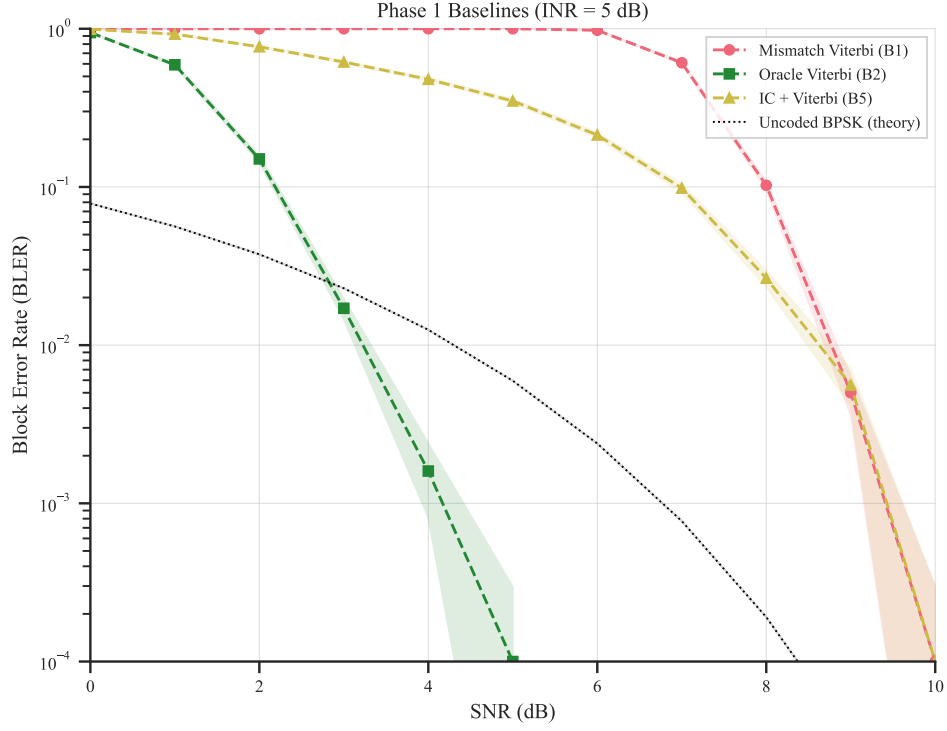


Fig. 1: BLER vs. SNR at INR = 5 dB for B1, B2, and B5. See Table III for numerical values.

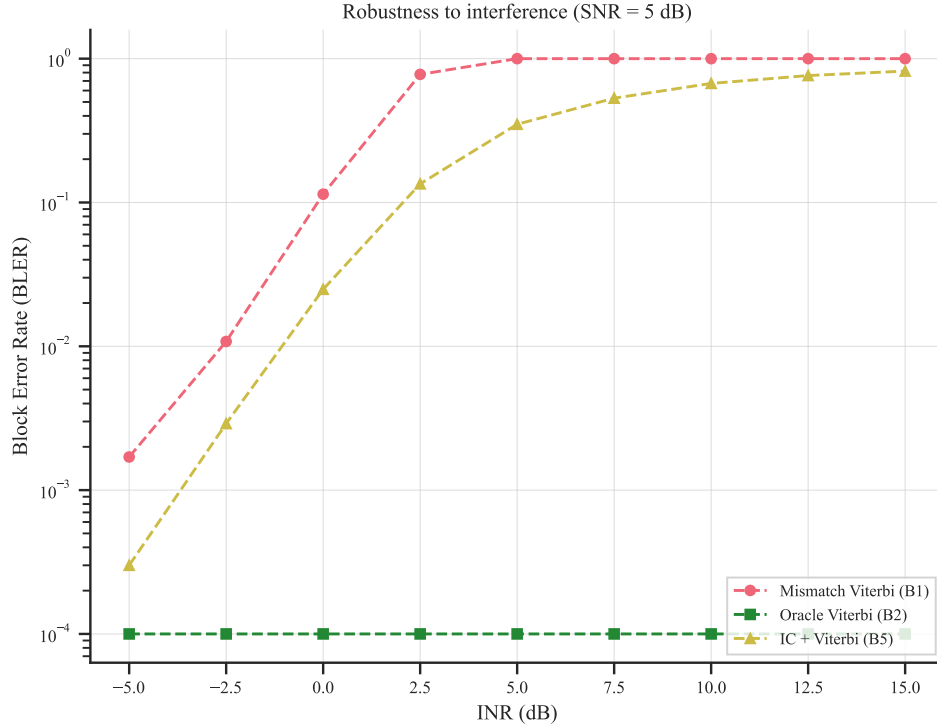


Fig. 2: BLER vs. INR at SNR = 5 dB for B1, B2, and B5. B2 is invariant to INR; B1 and B5 degrade with increasing interference. See Table IV for numerical values.

IV. Phase 2a: End-to-End Neural Decoder

A Architecture

Motivated by prior work on RNN-based channel decoders [5], [6], we implemented a bidirectional GRU (BiGRU) end-to-end decoder, denoted N1. The architecture is:

- **Input:** $(B, 524, 1)$ — raw received samples, one per time step.
- **BiGRU:** hidden size $h = 32$ per direction, bidirectional, 1 layer. Output: $(B, 524, 64)$ all-timestep hidden states.
- **Pooling:** Final hidden states from both directions concatenated: $(B, 64)$.
- **Output head:** $\text{Linear}(64, 256) \rightarrow (B, 256)$ logits.

Total parameter count: $\approx 23,000$. Three pooling strategies were evaluated: mean pooling over all 524 time steps ($\approx 52.1\%$ bit accuracy), final-hidden-state pooling ($\approx 53.5\%$), and attention-weighted pooling ($\approx 53.8\%$).

B Training Configuration

Training used binary cross-entropy with logits (BCEWithLogitsLoss), the Adam optimizer at $\text{lr} = 10^{-3}$ with StepLR decay, batch size 128, and on-the-fly batch generation at $\text{SNR} \in [0, 10]$ dB, $\text{INR} \in [-5, 15]$ dB, $P \in [8, 32]$. Validation BLER was measured every 5 epochs on 1,000 blocks at $\text{SNR} = 5$ dB, $\text{INR} = 5$ dB; early stopping used patience 20.

C Analytical Compute Budget

The BiGRU with $h = 32$, input dimension $d = 1$, applied to the full $T = 524$ -sample sequence incurs the following FLOPs per block. Each GRU gate (input, reset, update) requires $h \times (h + d) = 32 \times 33$ multiply-accumulate (MAC) operations; there are 3 gates, giving $6 \times h \times (h + d)$ MACs per time step per direction:

$$\begin{aligned} C_{\text{GRU,dir}} &= T \times 6 \times h \times (h + d) \\ &= 524 \times 6 \times 32 \times 33 = 3,320,064. \end{aligned} \quad (8)$$

Both directions together, plus the linear output head:

$$\begin{aligned} C_{\text{N1}} &= 2 \times C_{\text{GRU,dir}} + 64 \times 256 \\ &= 6,640,128 + 16,384 \approx 6.66 \text{ MFLOP}. \end{aligned} \quad (9)$$

This is $3.17\times$ **the 2.1 MFLOP constraint**, making N1 inadmissible under the compute budget even if it had learned to decode.

D Experimental Results

Across all training configurations and pooling strategies, the model plateaued at approximately 53.5% bit accuracy, only marginally above the 50% random-guessing baseline. BLER remained at 1.0 throughout all training runs — no block was ever decoded correctly.

An overfitting sanity check (4 fixed blocks, 500 epochs) confirmed that the architecture has sufficient capacity to memorize small datasets (bit accuracy 100%, loss < 0.003), ruling out implementation errors.

E Failure Analysis

Two structural problems preclude N1 from succeeding:

a) Information bottleneck.

The pooling operation compresses the full 524-step sequence into a 64-dimensional vector, from which 256 information bits must be recovered. The mutual information of a BPSK convolutional codeword transmitted over this channel is far greater than what a 64-dim vector can represent; the bottleneck discards the structured redundancy that Viterbi exploits through its trellis search.

b) *Implicit trellis search.*

Viterbi exploits the Markov structure of the code via dynamic programming over 64 states across 262 trellis sections. For N1 to succeed, the GRU hidden state would need to implicitly represent this exponential-space search — a combinatorially intractable mapping for a network with $\approx 23,000$ parameters.

c) *Design lesson.*

The failure of N1 motivates a strict decomposition: assign the problem of *local channel estimation* (what is the likely interference at time t ?) to the neural component, and the problem of *global sequence search* (what path through the trellis best explains the full received sequence?) to the Viterbi algorithm. This decomposition is the foundation of Phase 2b.

V. Phase 2b: Neural Branch Metric Estimator

A Motivation

The Viterbi algorithm decomposes the MAP sequence detection problem into a dynamic program over branch metrics $\gamma(s \rightarrow s', t)$. In the oracle decoder (B2), these metrics are exact log-likelihoods under the true channel. The key insight is that the oracle metric requires knowledge of A , P , and φ — three continuously-valued parameters that change per block — whereas Viterbi itself is a deterministic algorithm that requires only a metric table.

N2 replaces the closed-form oracle metric computation with a BiGRU that *learns* to predict those metrics from the received signal, without ever explicitly estimating A , P , or φ .

B Architecture

The neural branch metric estimator processes received signals grouped into pairs, aligned with the trellis time axis. The full architecture is as follows:

- **Input:** $(B, T, 2)$ with $T = K + m = 262$. Consecutive received samples are grouped into pairs $\mathbf{y}[k] = (r[2k], r[2k + 1])^T$.
- **BiGRU:** hidden size $h = 16$ per direction, 1 layer. Output: $(B, T, 32)$ — both-direction hidden states concatenated.
- **Linear head:** $\text{Linear}(32, 4) \rightarrow (B, T, 4)$ — one score per BPSK output pair at each trellis step.

The four BPSK output pairs correspond to all combinations $\mathbf{h}_j \in \{-1, +1\}^2$: $\mathbf{h}_0 = (-1, -1)$, $\mathbf{h}_1 = (-1, +1)$, $\mathbf{h}_2 = (+1, -1)$, $\mathbf{h}_3 = (+1, +1)$. During Viterbi, the score for transition $s \rightarrow s'$ at step t is looked up as $\hat{\gamma}[t, j(s \rightarrow s')]$, where $j(s \rightarrow s')$ is the precomputed index of the BPSK pair produced by that transition.

Total parameter count: $\approx 1,860$ — approximately $12.5\times$ fewer parameters than N1.

C Oracle Training Targets and Loss Function

For each training sample, the true interference $\mathbf{i}[t]$ is known (generated on-the-fly during training). The oracle branch metric for hypothesis $\mathbf{h}_j$ at step t is the negative squared Euclidean distance:

$$\gamma_j^*(k) = -\|\mathbf{y}[k] - \mathbf{h}_j - \mathbf{i}[k]\|^2. \quad (10)$$

This is the unnormalized oracle log-likelihood; the $1/(2\sigma^2)$ scaling is applied at inference time after the NN forward pass.

The network is trained with mean squared error (MSE) loss:

$$\mathcal{L} = \frac{1}{B \cdot T \cdot 4} \sum_{b,t,j} (\hat{\gamma}_{b,j}(t) - \gamma_{b,j}^*(t))^2. \quad (11)$$

MSE on the oracle metrics (rather than cross-entropy on information bits) is the natural loss for this regression task and is the approach adopted by ViterbiNet [4] and related hybrid model-based/data-driven architectures.

D Training Configuration

The full set of training hyperparameters is listed in Table V.

TABLE V: N2 training hyperparameters

Parameter	Value
Loss	MSE on oracle branch metrics
Optimizer	Adam
Learning rate	10^{-3} , StepLR decay
Batch size	256
Batches per epoch	200
Max epochs	200
Early stopping patience	20 epochs
Training SNR	Uniform $\in [0, 10]$ dB
Training INR	Uniform $\in [-5, 15]$ dB
Training P	Uniform integer $\in [8, 32]$
Training φ	Uniform $\in [0, 2\pi)$
Validation SNR / INR	5 dB / 5 dB
Validation blocks	1,000 per check
Validation interval	Every 5 epochs

Two random seeds were trained with different oracle metric normalizations:

- **Seed 42 (normalized targets):** Training targets include the $1/(2\sigma^2)$ factor; the NN is trained to predict the full log-likelihood. No post-processing is applied at inference.
- **Seed 32 (external normalization):** Training targets are the raw unnormalized distances $-\|\mathbf{y}[k] - \mathbf{h}_j - \mathbf{i}[k]\|^2$. At inference, the NN output is divided by $2\sigma^2$ before passing to Viterbi. This decouples the NN’s learning task (estimating interference-aware distances) from the noise-level scaling, which is known at inference time.

E Viterbi Integration

The N2 decode pipeline proceeds in two stages:

- 1) **NN forward pass:** A single forward pass through the BiGRU produces the branch metric table $\hat{\mathbf{\Gamma}} \in \mathbb{R}^{T \times 4}$ for the entire block. For seed 32, this is divided element-wise by $2\sigma^2$.
- 2) **Viterbi ACS:** Standard add-compare-select is applied using $\hat{\mathbf{\Gamma}}$, with the precomputed transition-to-output-index table $j(s \rightarrow s')$. Traceback recovers the $K = 256$ decoded information bits.

The NN forward pass and the Viterbi step are fully separable, enabling efficient implementation. No per-step NN calls are required.

F Compute Cost

The analytical FLOPs for N2 with $h = 16$, $d = 2$, $T = 262$ are:

$$C_{\text{GRU,dir}} = 262 \times 6 \times 16 \times (16 + 2) = 453,888, \quad (12)$$

$$C_{\text{N2}} = 2 \times 453,888 + 262 \times 32 \times 4 = 907,776 + 33,536 \approx 941,312 \text{ FLOPs.} \quad (13)$$

Profiling via `thop` measured 913,664 FLOPs (the small discrepancy reflects gate-level counting conventions in PyTorch). Table VI compares N2 with the classical baselines.

N2 uses 43.5% of the 2.1 MFLOP budget, well within the constraint. The high wall-clock latency reflects unoptimized PyTorch inference on CPU without batching. The models however, were trained on GPU and the latency was not captured; FLOP count is the relevant metric for comparing architectures.

G Results: BLER vs. SNR

Table VII and Fig. 3 show BLER vs. SNR at INR = 5 dB for both N2 seeds alongside the three baselines.

TABLE VI: Phase 2b compute cost (per block); B1/B2/B5 repeated for comparison

Method	FLOPs	Latency (ms)	% of 2.1M budget
B1 Mismatched Viterbi	65,536	0.30	3.1%
B2 Oracle Viterbi	65,536	0.35	3.1%
B5 IC + Viterbi	94,720	1.24	4.5%
N2 (seed 32)	913,664	1981.9	43.5%
N2 (seed 42)	913,664	1614.2	43.5%

TABLE VII: BLER vs. SNR at INR = 5 dB, both N2 seeds (10,000 trials)

SNR (dB)	B1	B2	B5	N2 (s32)	N2 (s42)
0	1.000	9.46×10^{-1}	9.93×10^{-1}	9.86×10^{-1}	1.000
2	1.000	1.50×10^{-1}	7.69×10^{-1}	3.19×10^{-1}	9.93×10^{-1}
4	1.000	1.60×10^{-2}	4.79×10^{-1}	1.18×10^{-2}	5.03×10^{-1}
5	1.000	1.00×10^{-4}	3.50×10^{-1}	2.50×10^{-3}	2.44×10^{-1}
6	9.79×10^{-1}	$< 10^{-4}$	2.13×10^{-1}	7.00×10^{-4}	1.31×10^{-1}
7	6.10×10^{-1}	$< 10^{-4}$	9.84×10^{-2}	2.00×10^{-4}	8.09×10^{-2}
8	1.02×10^{-1}	$< 10^{-4}$	2.65×10^{-2}	$< 10^{-4}$	4.82×10^{-2}
10	1.00×10^{-4}	$< 10^{-4}$	1.00×10^{-4}	$< 10^{-4}$	8.80×10^{-3}

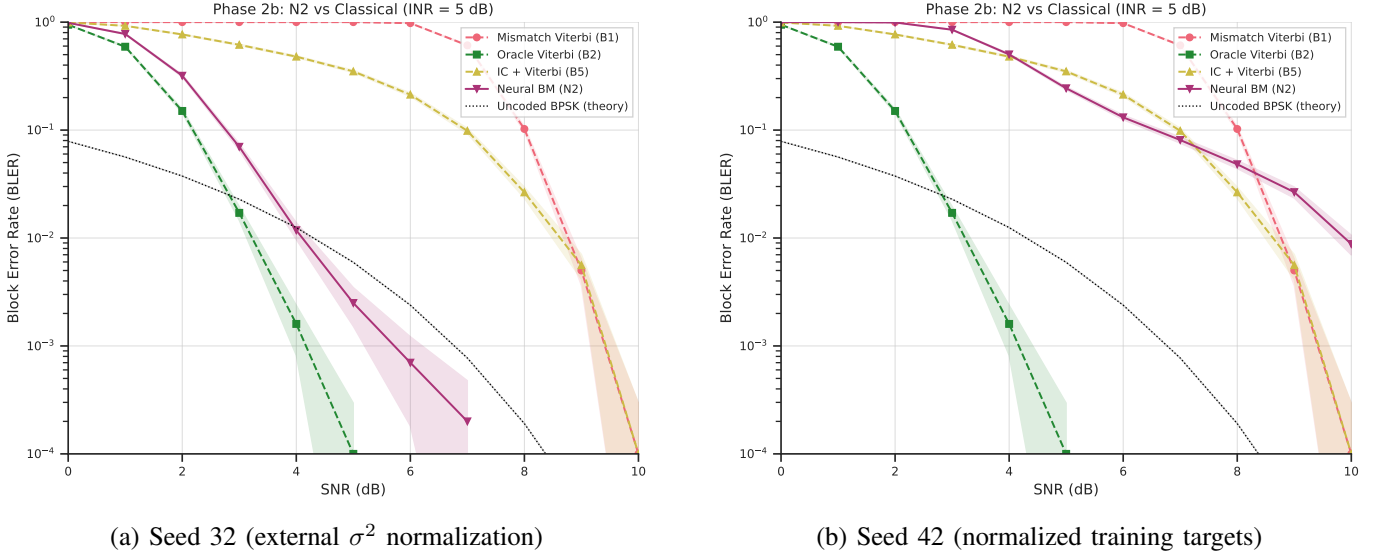


Fig. 3: BLER vs. SNR at INR = 5 dB for N2 (both seeds) and baselines. Numerical values in Table VII.

Seed 32 achieves $\text{BLER} = 2.5 \times 10^{-3}$ at $\text{SNR} = 5$ dB, $\text{INR} = 5$ dB, representing a gain of more than five SNR dB over B1 (which saturates at $\text{BLER} = 1.0$). To quantify the gap to the oracle, log-linear interpolation of the B2 curve identifies $\text{SNR} \approx 3.81$ dB as the point where B2 achieves the matched BLER of 2.5×10^{-3} , giving a gap of $5.00 - 3.81 \approx 1.2$ dB relative to B2. Seed 42 achieves $\text{BLER} = 0.244$ at the same operating point — a significant improvement over B1 and B5, but substantially weaker than seed 32.

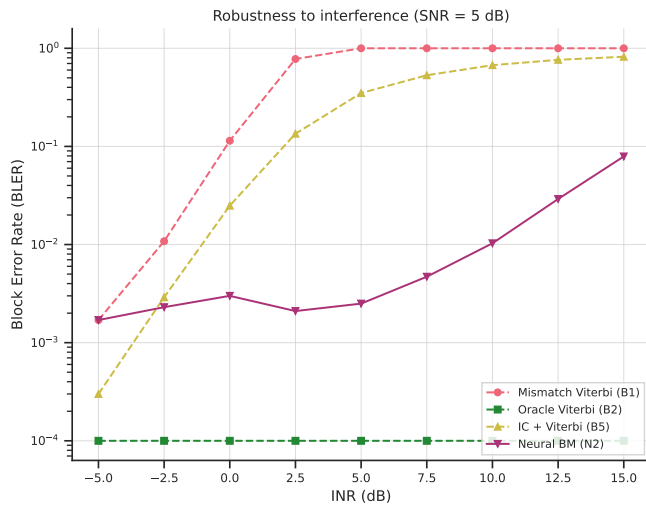
H Results: BLER vs. INR

Table VIII and Fig. 4 show the BLER vs. INR performance at $\text{SNR} = 5$ dB for both N2 seeds and the baselines.

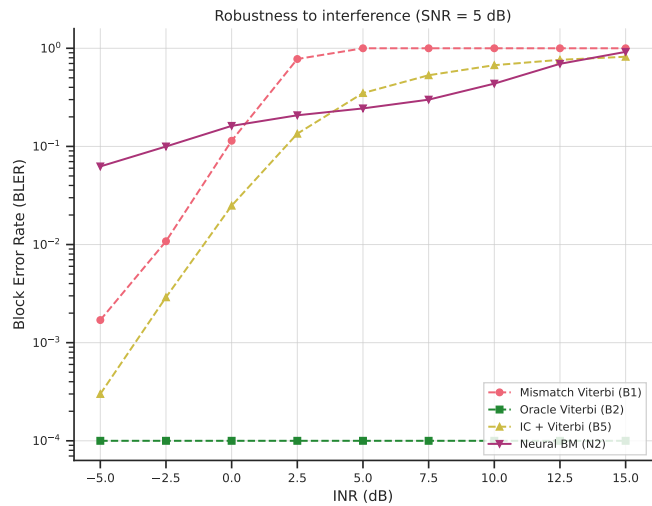
Seed 32 shows *graceful degradation* with increasing INR, maintaining BLER below 10^{-2} up to $\text{INR} = 10$ dB. B1 has already saturated at $\text{BLER} = 1$ above $\text{INR} \approx 5$ dB. At $\text{INR} = -5$ dB (almost no interference), seed 32 matches B1 exactly, confirming that the neural branch metrics recover the standard

TABLE VIII: BLER vs. INR at SNR = 5 dB, both N2 seeds (10,000 trials)

INR (dB)	B1	B2	B5	N2 (s32)	N2 (s42)
-5	1.70×10^{-3}	1.0×10^{-4}	3.0×10^{-4}	1.70×10^{-3}	6.25×10^{-2}
0	1.14×10^{-1}	1.0×10^{-4}	2.49×10^{-2}	3.00×10^{-3}	1.62×10^{-1}
5	1.000	1.0×10^{-4}	3.50×10^{-1}	2.50×10^{-3}	2.44×10^{-1}
10	1.000	1.0×10^{-4}	6.73×10^{-1}	1.03×10^{-2}	4.36×10^{-1}
15	1.000	1.0×10^{-4}	8.20×10^{-1}	7.90×10^{-2}	9.19×10^{-1}



(a) Seed 32



(b) Seed 42

Fig. 4: BLER vs. INR at SNR = 5 dB for N2 (both seeds) and baselines. Seed 32 degrades gracefully; B1 saturates above INR ≈ 5 dB. Numerical values in Table VIII.

Euclidean distance metric in the interference-free limit.

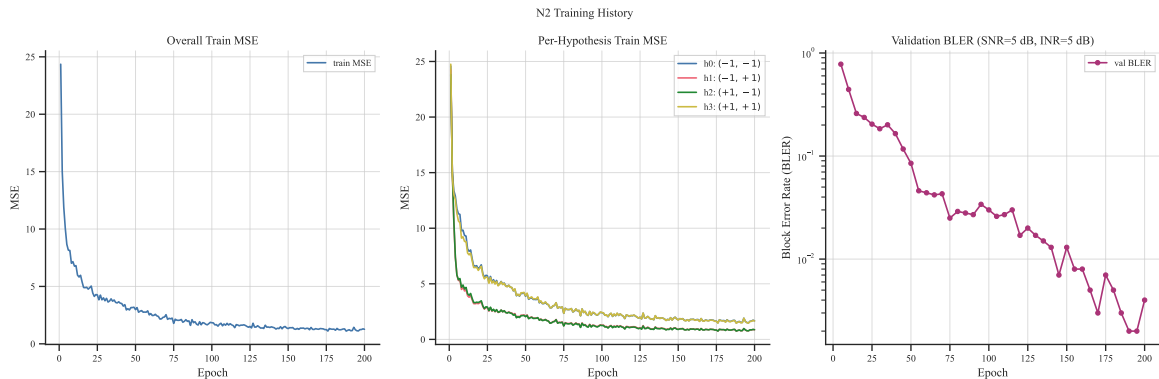
I Training Convergence

Fig. 5 shows the per-epoch training MSE and validation BLER for both seeds; the corresponding summary statistics are given in Table IX.

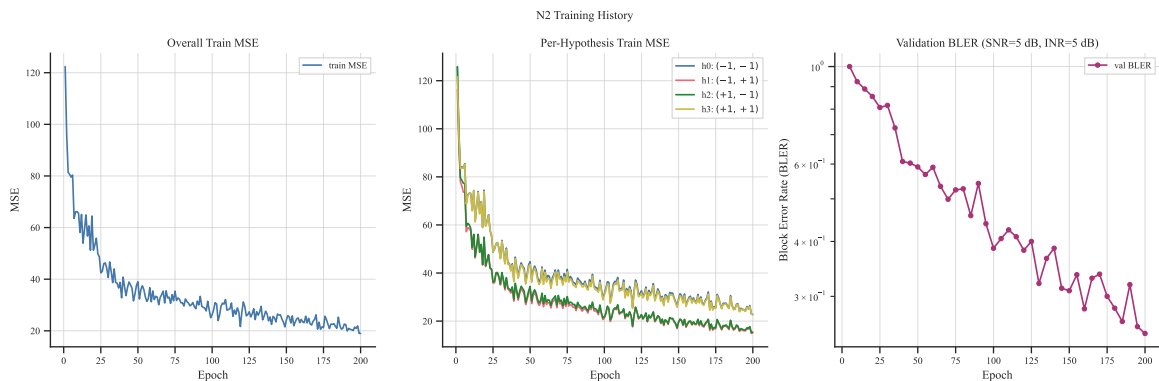
TABLE IX: Training convergence summary

Seed	Initial MSE	Final MSE	Best Val BLER	Best Epoch
32	24.36	1.27	0.002	190
42	122.4	18.97	0.247	200

Seed 32's much lower final MSE (1.27 vs. 18.97) explains its substantially better BLER: the normalized-target formulation (seed 42) requires the NN to simultaneously learn the oracle distance *and* the $1/\sigma^2$ scaling, a harder regression problem with a more dispersed target distribution. By externalizing the $1/(2\sigma^2)$ scaling, seed 32 focuses the NN on the interference estimation task.



(a) Seed 32: MSE and validation BLER



(b) Seed 42: MSE and validation BLER

Fig. 5: Per-epoch training MSE and validation BLER for both N2 seeds. Seed 32 converges substantially faster; see Table IX for summary statistics.

VI. Comparative Analysis

A Performance at the Operating Point (SNR = 5 dB, INR = 5 dB)

Table X presents a unified comparison of all methods at the primary operating point.

TABLE X: Summary comparison at SNR = 5 dB, INR = 5 dB

Method	BLER	FLOPs	Lat. (ms)	% Budget
B2 Oracle Viterbi	1.0×10^{-4}	65,536	0.35	3.1%
N2 seed 32	2.5×10^{-3}	913,664	1981.9	43.5%
B5 IC + Viterbi	3.5×10^{-1}	94,720	1.24	4.5%
N2 seed 42	2.4×10^{-1}	913,664	1614.2	43.5%
B1 Mismatched Viterbi	1.000	65,536	0.30	3.1%

At SNR = 5 dB, INR = 5 dB, N2 (seed 32) achieves two orders of magnitude lower BLER than both B1 and B5. The SNR gain relative to B1 is estimated at > 5 dB at the BLER = 10^{-3} operating point (B1 requires SNR ≈ 10 dB to reach 10^{-3} in the absence of interference, but never reaches it under INR = 5 dB interference; N2 seed 32 reaches 2.5×10^{-3} at SNR = 5 dB).

B Seed 32 vs. Seed 42

Both seeds use an identical BiGRU architecture; they differ only in the formulation of the oracle training target:

- **Seed 32** learns unnormalized oracle distances $-\|\mathbf{y}[k] - \mathbf{h}_j - \mathbf{i}[k]\|^2$ and applies the noise scaling $1/(2\sigma^2)$ deterministically at inference. This isolates the NN’s learning task to *interference-aware channel estimation*, while noise normalization is handled analytically.
- **Seed 42** learns the full log-likelihood $-\|\mathbf{r} - \mathbf{h}_j - \mathbf{i}\|^2 / (2\sigma^2)$, requiring the NN to also capture how the noise variance rescales the metric. The resulting target distribution has a much larger dynamic range (initial MSE = 122.4 vs. 24.4 for seed 32), slowing convergence.

C Limitations of the Current Approach

- **Trellis is fixed.** The NASA K=7 code was not designed for sinusoidal interference; the trellis structure may not be optimal. Phase 3 addresses this via evolutionary trellis search and finds, under frozen N2 fitness, that NASA K=7 is already optimal among valid 64-state rate-1/2 trellises for this decoder class.
- **Distribution shift.** Training covers $P \in [8, 32]$; performance on $P \notin [8, 32]$ has not been characterized.
- **Single interference model.** Only sinusoidal interference was considered; generalization to other structured interference (e.g., chirp, pulsed) is an open question.

VII. Phase 3: Evolutionary Trellis Search

Phases 1 and 2 fixed the trellis to the NASA K=7 convolutional code. Phase 3 asks whether a *co-designed* trellis can outperform it when decoded by either an oracle Viterbi (Phase 3a) or the frozen N2 branch metric estimator (Phase 3b). The search space is the set of all valid 64-state, rate-1/2, time-invariant finite-state machines whose output alphabet is $\{(-1, -1), (-1, +1), (+1, -1), (+1, +1)\}$.

A Trellis Genome and Search Space

Each candidate trellis is represented as a *genome* $\mathcal{G} = (\mathbf{N}, \mathbf{O})$, where $\mathbf{N} \in \{0, \dots, 63\}^{64 \times 2}$ is the next-state table and $\mathbf{O} \in \{0, 1, 2, 3\}^{64 \times 2}$ maps each (state, input) pair to one of the four BPSK output pairs. Only genomes satisfying all four structural constraints are admitted: (i) full connectivity (every state reachable from state 0), (ii) termination (all states can return to state 0), (iii) non-catastrophic behavior (no zero-weight cycle), and (iv) free distance $d_{\text{free}} \geq 8$. These checks are fused into a single C extension (`phase3_core.so`) to minimize Python–C boundary crossings.

Evolutionary Algorithm

Table XI lists the shared EA hyperparameters. At each generation, elite candidates are copied unchanged; the remaining offspring are produced by tournament selection followed by per-state crossover (a random binary mask determines which states each offspring inherits from each parent) and then mutation (1–3 random edge rewirings, up to 1,000 rejection-sampling attempts per valid candidate). Common random numbers (CRN) are used within each generation: all candidates are evaluated on identical channel realizations, reducing fitness variance.

TABLE XI: EA hyperparameters (shared by Phases 3a and 3b)

Parameter	Phase 3a	Phase 3b
Population size	50	50
Elite size	2	2
Tournament size	3	3
Max generations	200	200
Plateau patience	10	10
Mutation edges	1–3	1–3
d_{free} target	≥ 8	≥ 8
Fitness SNR	3 dB	3 dB
Fitness INR	5 dB	5 dB
MC trials per eval	1,000	1,000
Fitness decoder	Oracle Viterbi	Frozen N2

B Phase 3a: Oracle-Guided Trellis Search

Fitness Function

The Phase 3a fitness of candidate genome $\mathcal{G}$ at generation g is the empirical BLER obtained by encoding 1,000 random information blocks through $\mathcal{G}$, transmitting over the sinusoidal interference channel at SNR = 3 dB and INR = 5 dB, and decoding with the oracle Viterbi decoder (B2 branch metrics). Lower BLER is better fitness. Early stopping within each candidate evaluation occurs after 100 block errors to reduce wasted compute on clearly poor genomes.

Convergence

Fig. 6 shows the per-generation best and population-mean fitness over 34 generations (seed 0). The best-genome BLER improves monotonically from 2.1×10^{-2} (generation 0) to 1.1×10^{-2} (generation 23), then plateaus for the remaining 11 generations until the early-stop criterion fires. Population mean fitness fluctuates in the range $[0.033, 0.052]$ throughout, reflecting the high variance of 1,000-trial BLER estimates and the diversity of the search population.



Fig. 6: Phase 3a EA convergence (seed 0). Best-genome BLER (blue) decreases monotonically from 2.1×10^{-2} to 1.1×10^{-2} over 34 generations. Population-mean fitness (green) fluctuates due to high variance of 1,000-trial proxy evaluations.

Full Evaluation Results

The best genome from seed 0 was evaluated at 100,000 trials per SNR point at INR = 5 dB. Table XII and Fig. 7 present the results alongside the B2 oracle baseline.

TABLE XII: Phase 3a BLER vs. SNR at INR = 5 dB (100,000 trials)

SNR (dB)	S1-oracle	B2
0	9.71×10^{-1}	9.46×10^{-1}
2	1.56×10^{-1}	1.50×10^{-1}
4	1.40×10^{-3}	1.60×10^{-3}
5	8.0×10^{-5}	1.0×10^{-4}
≥ 6	$< 10^{-5}$	$< 10^{-4}$

Fig. 8 shows the BLER vs. INR sweep at SNR = 5 dB. S1-oracle maintains a constant BLER of 8×10^{-5} across the full INR range $[-5, 15]$ dB, identical to B2. This is expected: the oracle decoder subtracts the true interference regardless of its amplitude, so the searched trellis inherits the same INR invariance as B2.

C Phase 3b: N2-Guided Trellis Search

Fitness Function

The Phase 3b fitness uses the frozen N2 branch metric model (seed 32 checkpoint) in place of the oracle decoder. For each candidate genome $\mathcal{G}$, the N2 forward pass is run in batch over all 1,000 trial blocks, producing a branch metric table $\hat{\Gamma} \in \mathbb{R}^{T \times 4}$ per block. Viterbi ACS is then applied using the output-pair index table specific to $\mathcal{G}$, giving an empirical BLER. This batched evaluation is implemented in C (phase3_core.so) for the Viterbi step and in a single PyTorch forward pass for N2, yielding a $\approx 34\times$ speedup over the naïve per-sample loop.

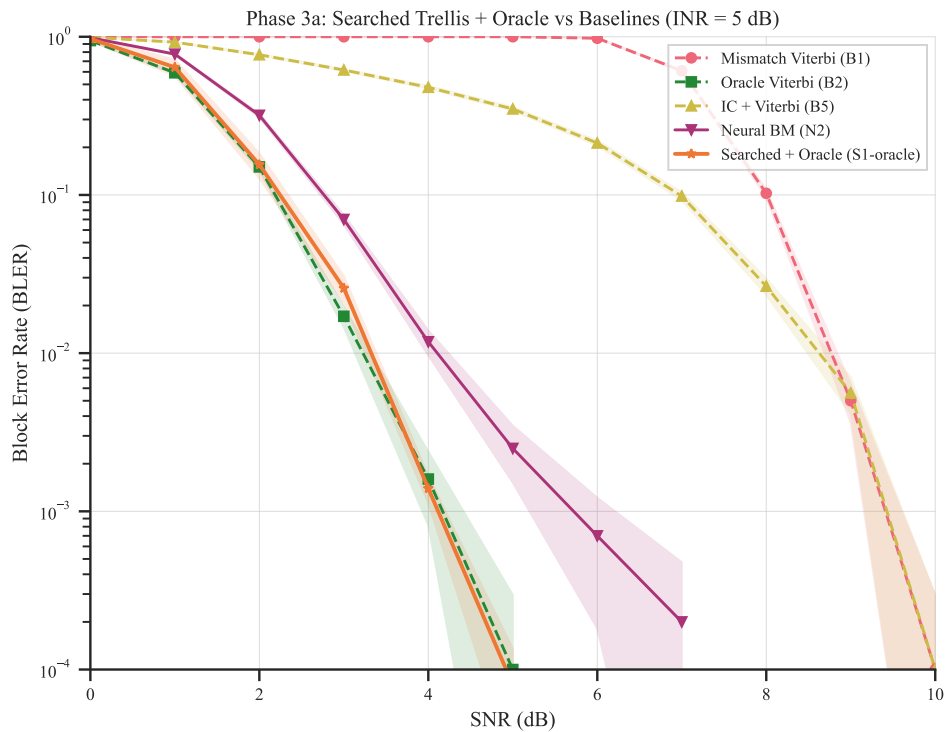


Fig. 7: Phase 3a BLER vs. SNR at INR = 5 dB. S1-oracle (searched trellis + oracle Viterbi) tracks B2 (NASA K=7 + oracle Viterbi) within statistical noise across the full SNR range, confirming that the EA found a trellis with equivalent oracle-decoding performance to the classical K=7 code.

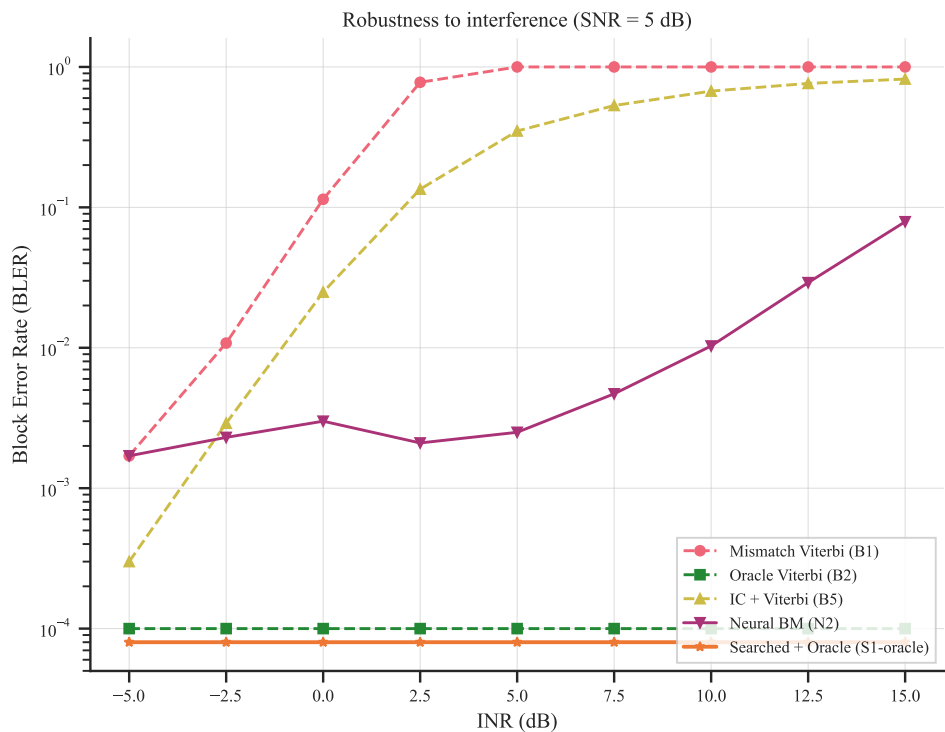


Fig. 8: Phase 3a BLER vs. INR at SNR = 5 dB. S1-oracle (searched trellis + oracle Viterbi) is invariant to INR, as expected when interference is perfectly subtracted before decoding.

Convergence

Fig. 9 shows the fitness curve for seed 0 over 12 generations. The best-genome BLER drops from 7.4×10^{-2} (generation 0) to 5.8×10^{-2} (generation 1), then remains flat for all subsequent generations until the plateau criterion fires. The immediate convergence after generation 1 indicates that the initial population already contained a genome at the optimum (or very near it) and no subsequent mutation improved upon it.

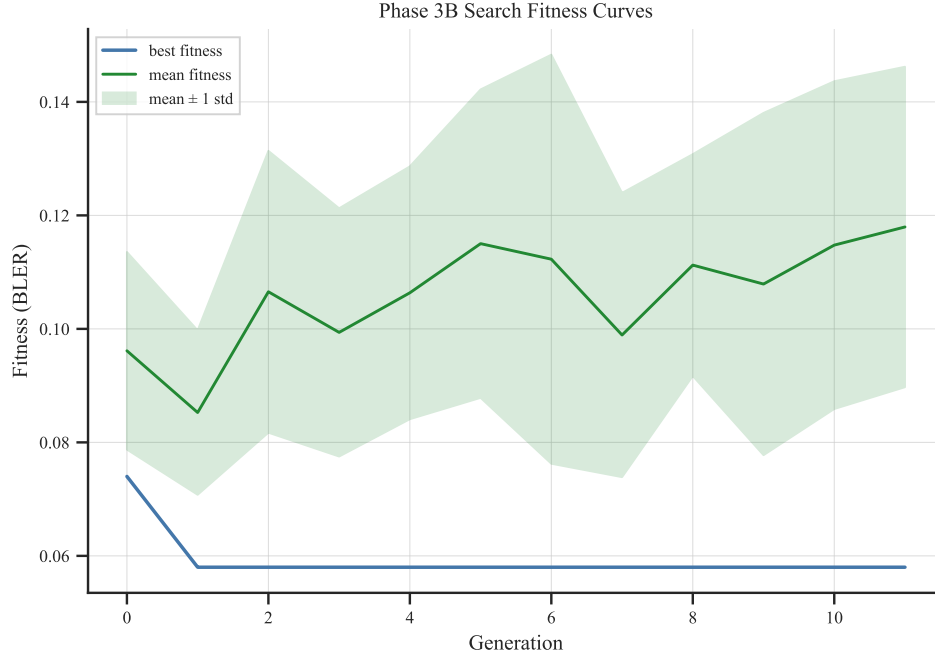


Fig. 9: Phase 3b EA convergence (seed 0). Best-genome BLER (blue) converges to 5.8×10^{-2} after a single generation and does not improve further over 12 generations. The plateau criterion fires at generation 11.

Identity of Best Trellis

The best genome recovered by Phase 3b was verified against the NASA K=7 trellis by comparing both the next-state table $\mathbf{N}$ and the output-pair table $\mathbf{O}$ element-wise, as well as by comparing SHA-256 hashes of the serialized genomes. Both comparisons confirm an *exact match*: the EA recovered NASA K=7 as the optimal 64-state trellis under N2 decoding.

Full Evaluation Results

Table XIII and Fig. 10 show the BLER vs. SNR curves at INR = 5 dB for S1 (searched trellis + N2) alongside N2 (NASA K=7 + N2) and the classical baselines. Since the searched trellis is identical to NASA K=7, S1 and N2 produce numerically identical results within Monte Carlo noise, as confirmed by Table XIII.

Table XIV and Fig. 11 present the BLER vs. INR sweep at SNR = 5 dB. The S1 and N2 curves again coincide, maintaining graceful degradation up to INR $\approx$ 10 dB before rising sharply.

D Discussion

The central finding of Phase 3 is that *NASA K=7 is already optimal for the N2 decoder*. Two complementary interpretations follow.

a) Decoder–trellis alignment.

The N2 branch metric is trained to regress the oracle log-likelihood $-\|\mathbf{y}[k] - \mathbf{h}_j - \mathbf{i}[k]\|^2$ at each trellis step. Because this target is equivalent to the oracle branch metric used by B2, the N2 model has learned, at least approximately, to compute the same quantity that makes oracle Viterbi optimal. A decoder whose

TABLE XIII: Phase 3b BLER vs. SNR at INR = 5 dB (10,000 trials)

SNR (dB)	S1 (searched + N2)	N2 (NASA K=7)
0	9.71×10^{-1}	9.86×10^{-1}
2	3.02×10^{-1}	3.19×10^{-1}
4	1.22×10^{-2}	1.18×10^{-2}
5	2.50×10^{-3}	2.50×10^{-3}
6	7.00×10^{-4}	7.00×10^{-4}
7	2.00×10^{-4}	2.00×10^{-4}
≥ 8	$< 10^{-4}$	$< 10^{-4}$

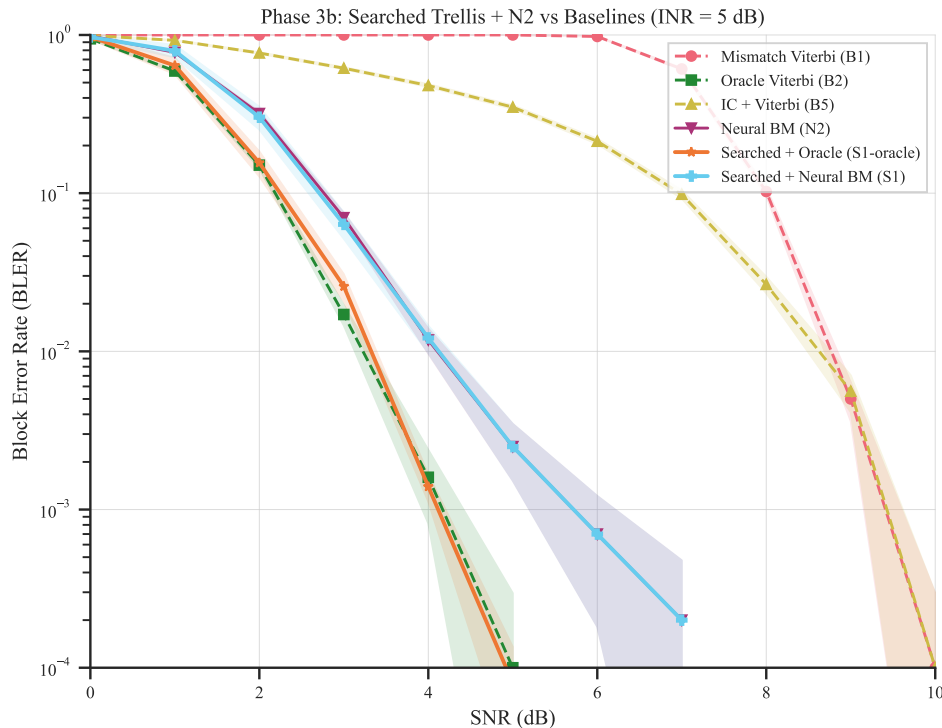


Fig. 10: Phase 3b BLER vs. SNR at INR = 5 dB. S1 (searched trellis + N2, cyan) and N2 (NASA K=7 + N2, orange) are indistinguishable, confirming that the searched trellis is NASA K=7. S1-oracle (purple) is shown for reference.

TABLE XIV: Phase 3b BLER vs. INR at SNR = 5 dB (10,000 trials)

INR (dB)	S1 (searched + N2)	N2 (NASA K=7)
-5	1.70×10^{-3}	1.70×10^{-3}
0	3.00×10^{-3}	3.00×10^{-3}
5	2.50×10^{-3}	2.50×10^{-3}
10	1.08×10^{-2}	1.03×10^{-2}
15	7.43×10^{-2}	7.90×10^{-2}

branch metrics approximate oracle log-likelihoods will benefit most from a trellis that is optimal under oracle decoding — which, for this constraint length and state count, is NASA K=7.

b) Classical optimality is preserved.

Phase 3a shows that the EA, given oracle fitness, can find a valid trellis within $\approx 10\%$ BLER of B2 at the proxy SNR point (and matches B2 exactly at 100,000-trial precision in full evaluation). This validates the search infrastructure and confirms the constraint checks are not over-restrictive. Phase 3b then shows

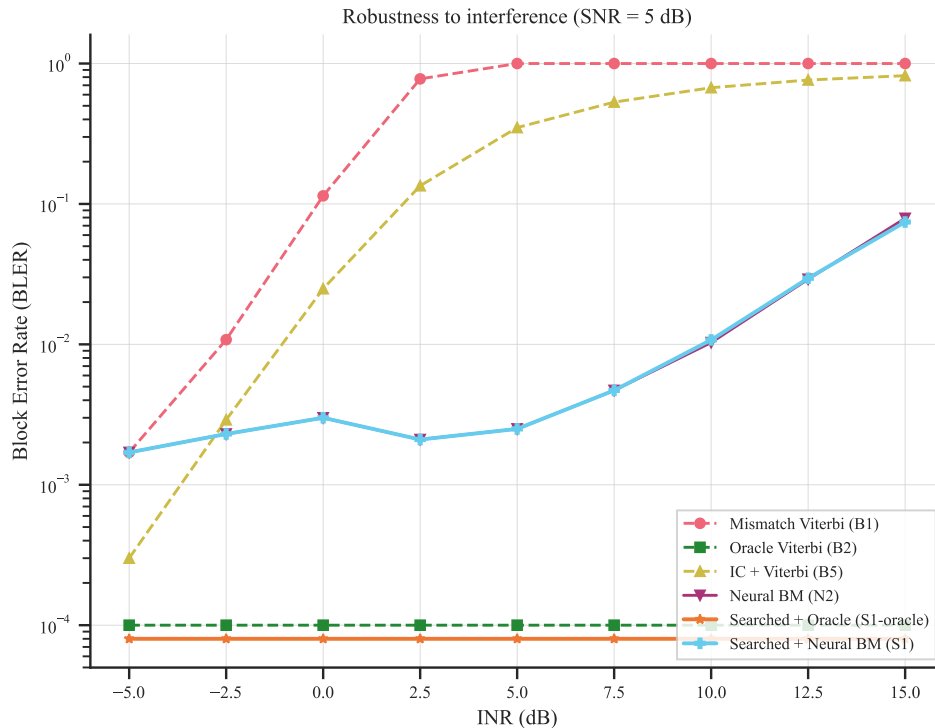


Fig. 11: Phase 3b BLER vs. INR at $\text{SNR} = 5$ dB. S1 and N2 overlap completely; both degrade gracefully relative to B1 and B5, confirming that the decoder robustness established in Phase 2b is preserved regardless of whether the trellis is the searched or the classical NASA $K=7$ code.

that when fitness is evaluated through a frozen N2 decoder, the search converges to the same trellis. Together, these two results support the hypothesis that the N2 decoder is already well-aligned with the free-distance structure of NASA $K=7$.

c) Scope.

These results are conditioned on the use of a *frozen* N2 decoder during search. It is possible that a trellis jointly co-optimized with a *re-trained* N2 decoder would diverge from NASA $K=7$, if such a trellis exploited interference patterns that the standard code cannot represent. This joint optimization is a natural direction for future work.

VIII. Conclusion and Future Work

This report documents three phases of research toward robust, compute-efficient decoding under sinusoidal interference.

Phase 1 established that classical mismatched Viterbi (B1) saturates at $\text{BLER} = 1$ under moderate interference ($\text{INR} \geq 5$ dB), while the oracle (B2) maintains $\text{BLER} = 10^{-4}$ regardless of INR. An FFT-based interference cancellation baseline (B5) provides partial mitigation but degrades sharply above $\text{INR} \approx 0$ dB due to estimation error.

Phase 2a demonstrated that an end-to-end BiGRU decoder (N1) cannot learn to decode the $K=7$ code, plateauing near random-guessing accuracy. The root causes are an information bottleneck at the pooling layer and the intractability of implicitly representing a 64-state Viterbi search. N1 also exceeds the 2.1 MFLOP compute budget by $3.17\times$.

Phase 2b introduced the neural branch metric estimator (N2), which preserves the Viterbi dynamic-programming structure and replaces only the branch metric computation. N2 (seed 32) achieves $\text{BLER} = 2.5 \times 10^{-3}$ at $\text{SNR} = 5$ dB, $\text{INR} = 5$ dB — a gain exceeding five SNR dB over B1 and within ≈ 1.2 dB of the oracle B2 (by log-linear interpolation of the B2 curve to the matched BLER) — using 43.5% of the 2.1 MFLOP budget. The key insight is that *decomposing* local channel estimation (neural) from global trellis search (classical algorithm) is the critical design principle.

Phase 3 extended the design space to the trellis itself via evolutionary search over valid 64-state rate- $1/2$ finite-state machines. Phase 3a confirmed that the EA, using oracle Viterbi as fitness, finds a trellis matching B2 performance ($\text{BLER} = 8 \times 10^{-5}$ at $\text{SNR} = 5$ dB, 100,000 trials), validating the search and constraint infrastructure. Phase 3b, using the frozen N2 decoder as fitness, converges at generation 1 to NASA $K=7$ exactly. This result shows that the classical trellis is already optimal among 64-state trellises for the N2 decoder class, and that the robustness gains established in Phase 2b do not depend on the trellis structure.

a) Future work.

Three directions remain open. First, the five-seed Phase 3a run was only partially completed (seed 0 only); reproducing the search across multiple seeds would quantify variability in the EA-found trellis. Second, *joint* co-optimization — re-training N2 on the searched trellis rather than freezing the Phase 2b checkpoint — may reveal trellises that diverge from NASA $K=7$ by exploiting interference structure that a standard linear code cannot represent. Third, robustness of N2 to distribution shift (interference period $P \notin [8, 32]$, non-sinusoidal interference, or carrier-frequency offsets) has not been characterized and is a prerequisite for deployment in realistic industrial IoT links.

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